

"Your Repo Is Fighting Your AI Agents"

2 min

70% of AI coding tasks fail — not because models are bad, but because **context** is wrong.

53%

Baseline task success

100%

AI-native repo success

1

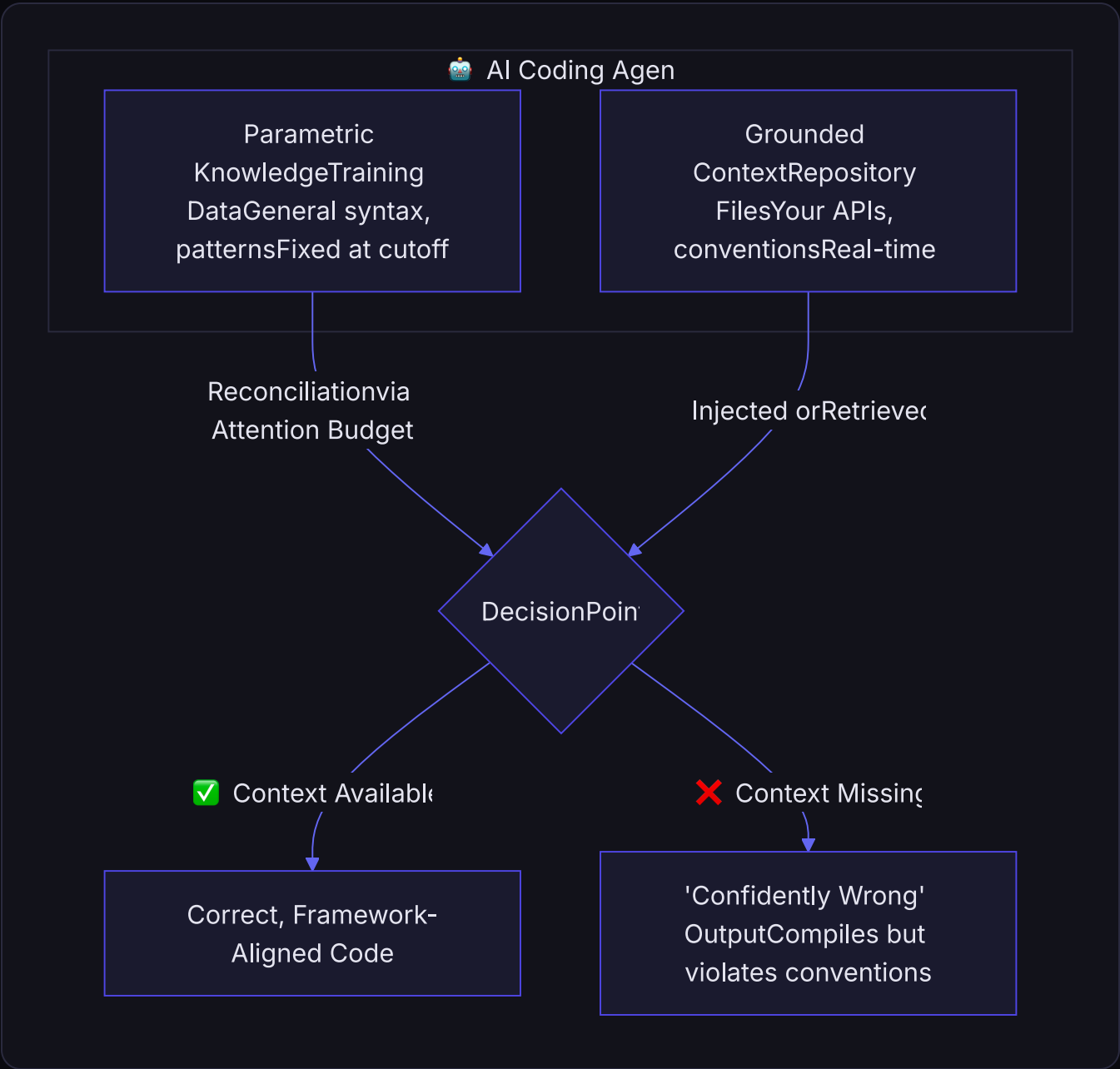
afternoon

Time to transform



Parametric vs. Grounded Knowledge 3 min

Every AI agent reconciles two knowledge sources via its attention budget. When grounded context is missing, parametric knowledge fills the gap — **confidently wrong**.



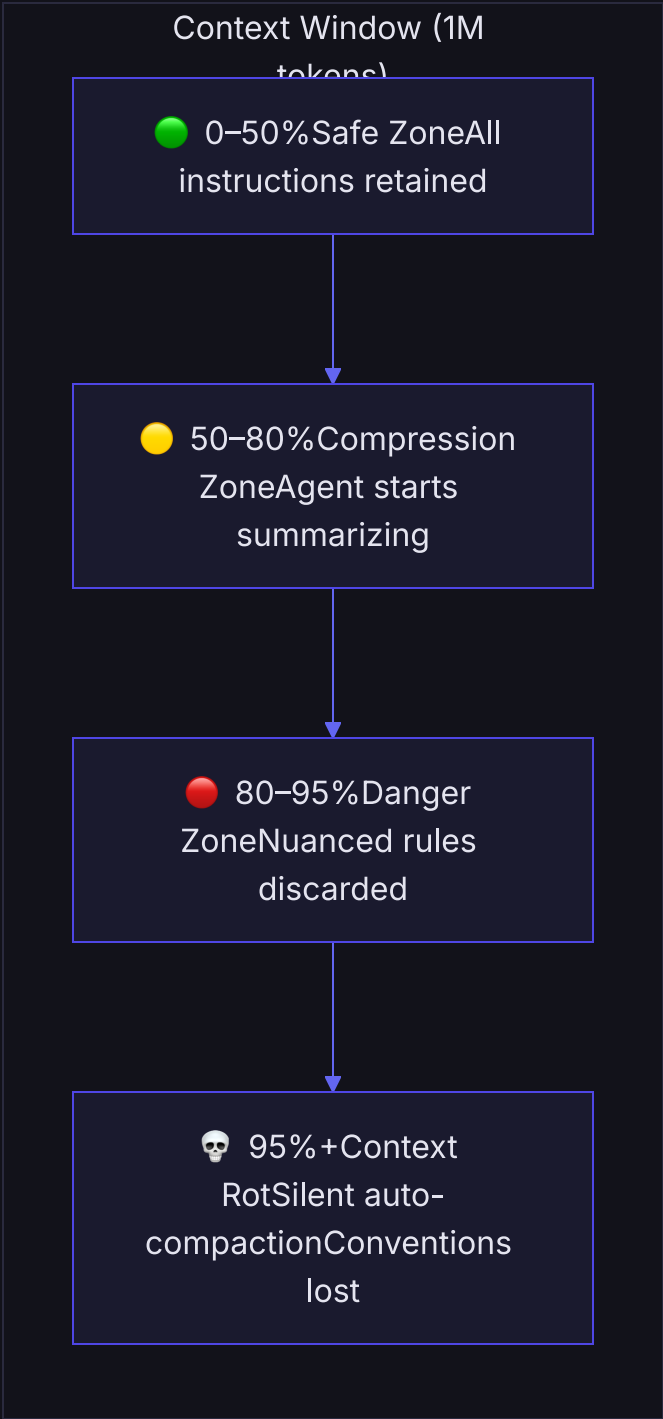
KEY INSIGHT

Vercel found agents skip framework docs **44% of the time** because they think they already know the answer.

The Silent Killer

2 min

Even million-token windows have a ticking time bomb. Agents silently auto-compact — your conventions are the first things discarded.



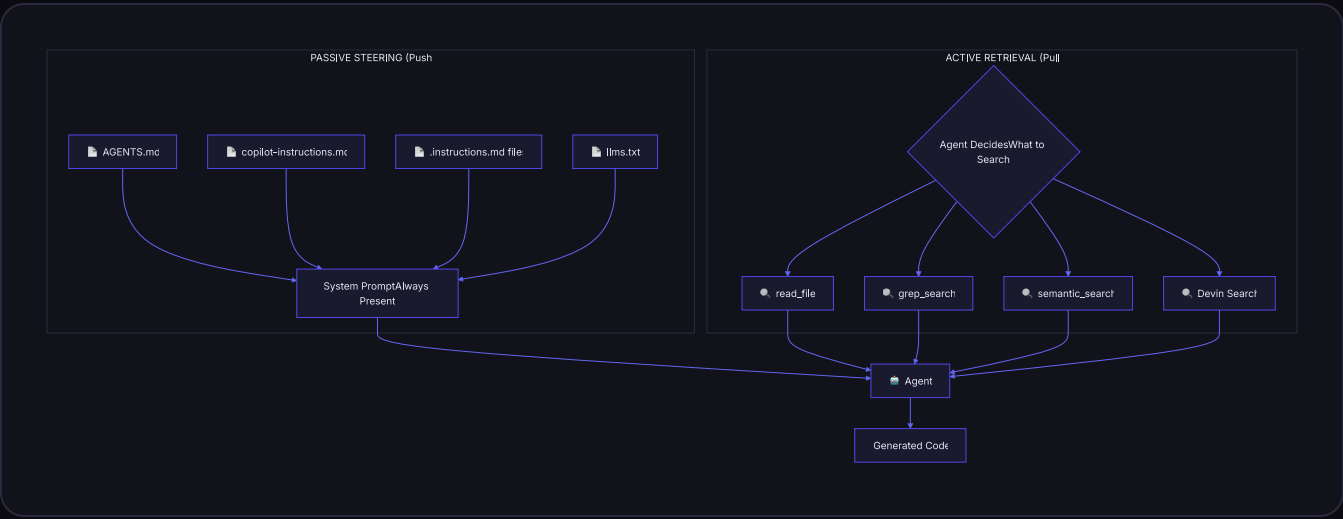
WHY THIS MATTERS

The code compiles. Tests might pass. But it silently violates your architecture. This is why we need **structured context**, not just throwing everything into the window.

Two Models of Context Delivery

3 min

Passive steering (push) vs. Active retrieval (pull). Passive wins reliability. Active wins scale.



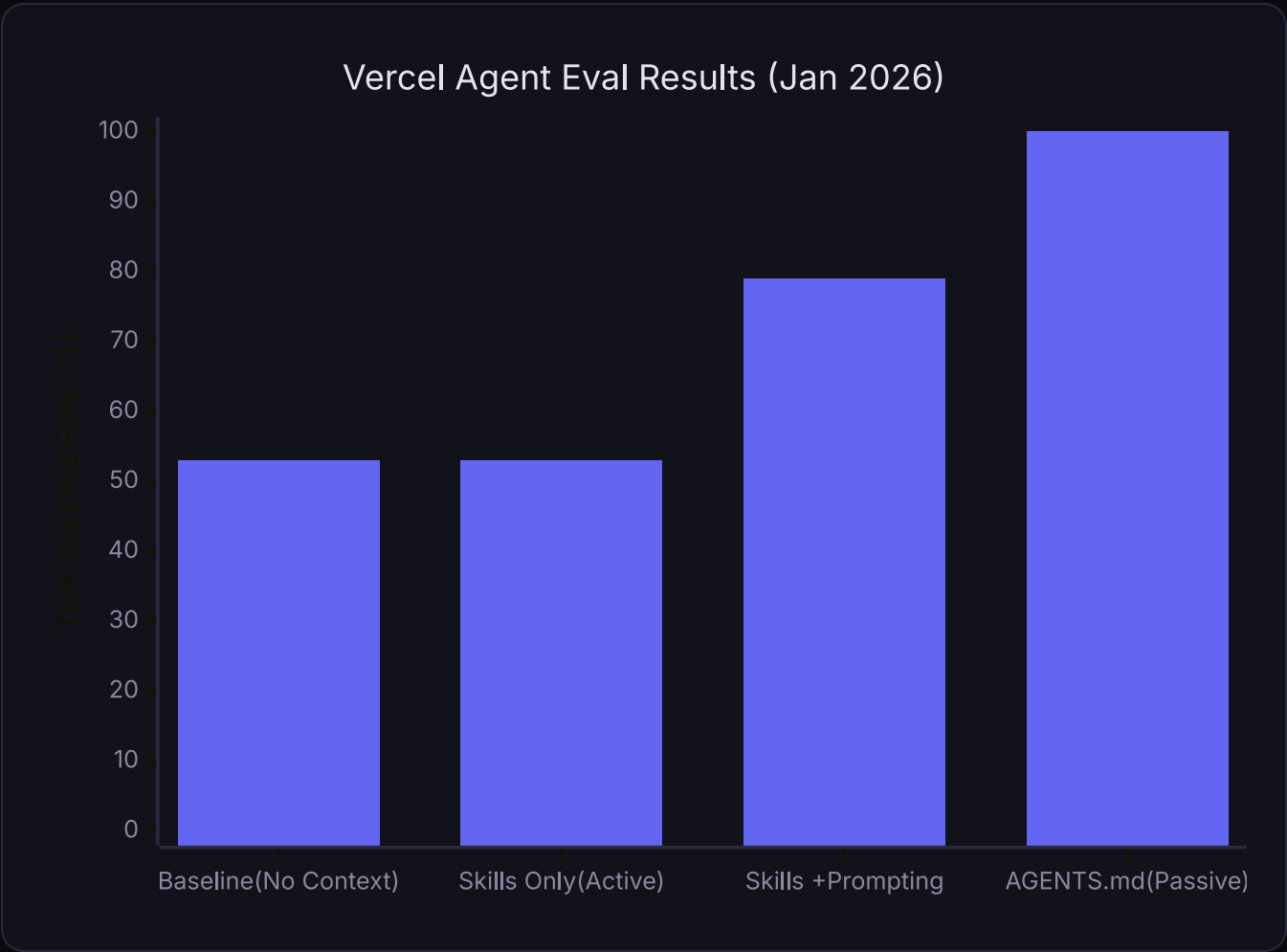
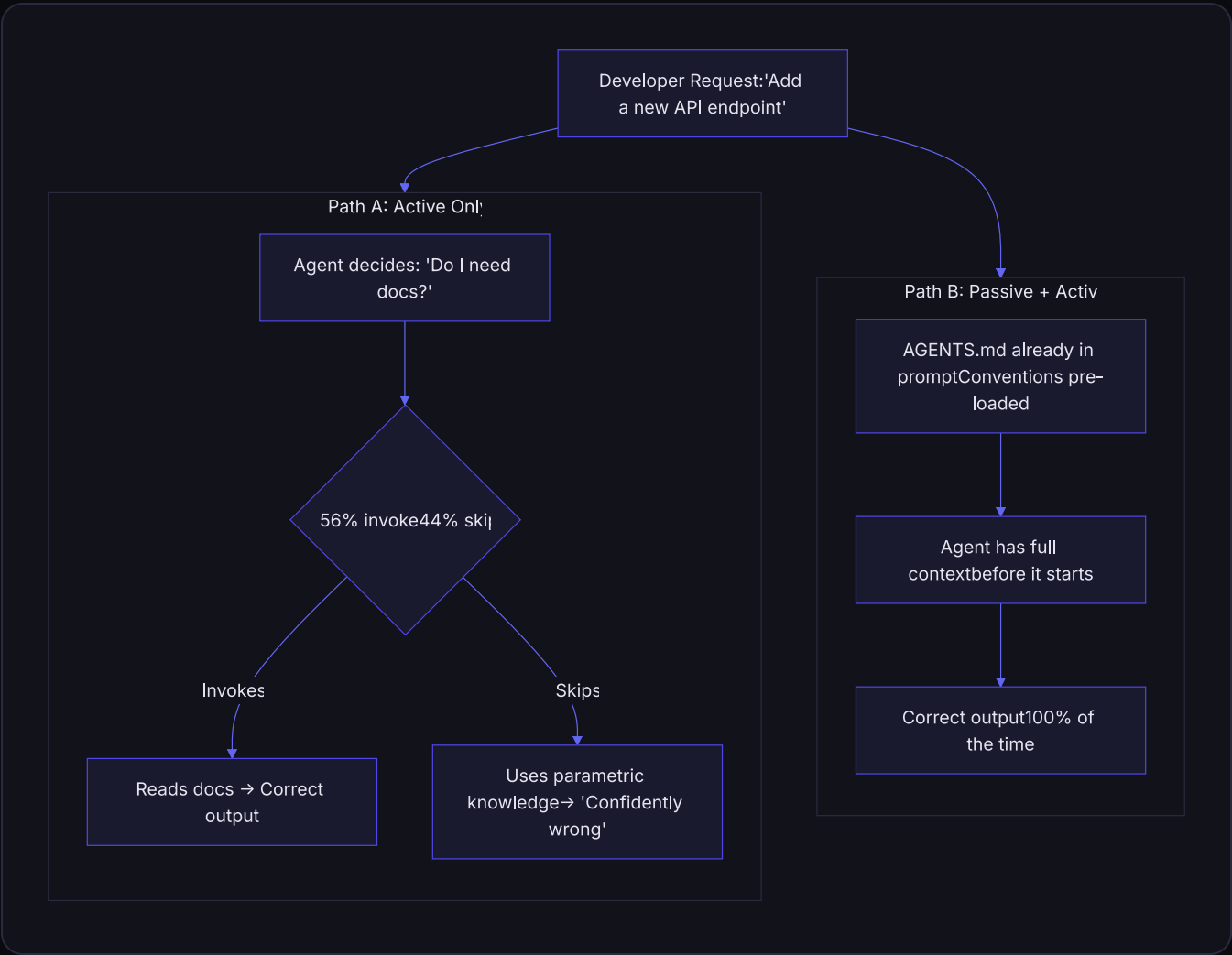
VERCEL'S KEY INSIGHT

Passive eliminates the Decision Gap. The agent can't skip what's already in its prompt.

Why Passive Wins

2 min

When agents decide whether to search, they skip docs 44% of the time.
Passive context removes the decision entirely.



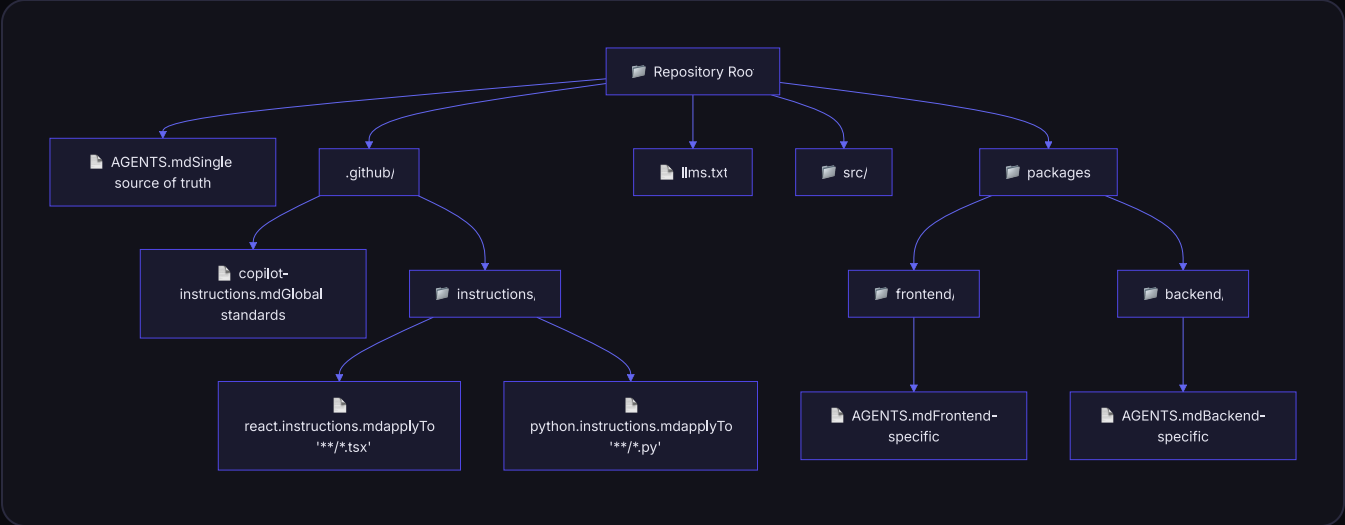
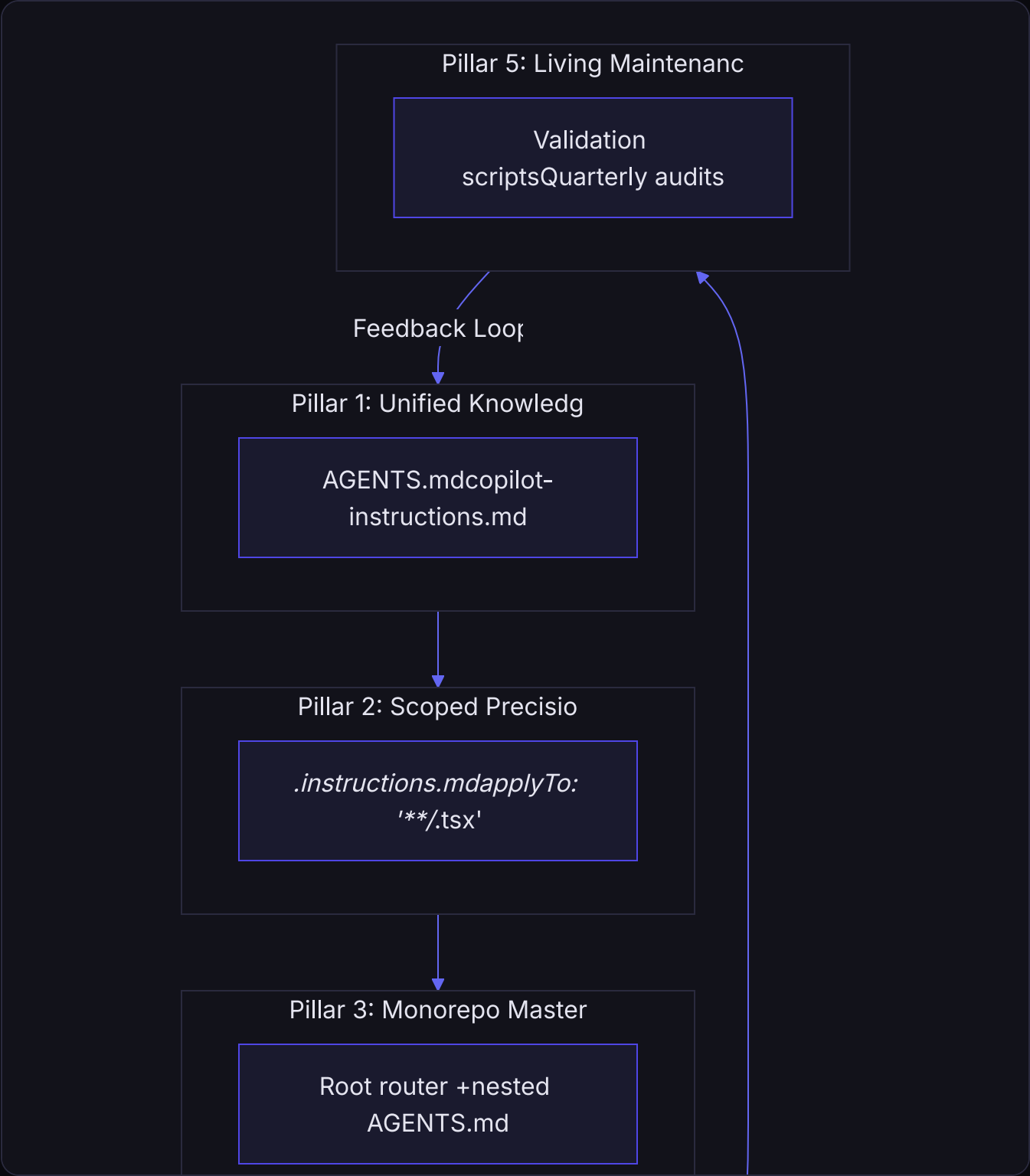
CRITICAL RESULT

A compressed **8KB AGENTS.md** achieved **100% success rate** — beating 40KB of full context (40KB full context = 53% success rate)

The Five Pillars of AI-Native Repos

3 min

Unified Knowledge → Scoped Precision → Monorepo Mastery → AI-Readable Docs → Living Maintenance



Anthropic's Progressive Disclosure 2 min

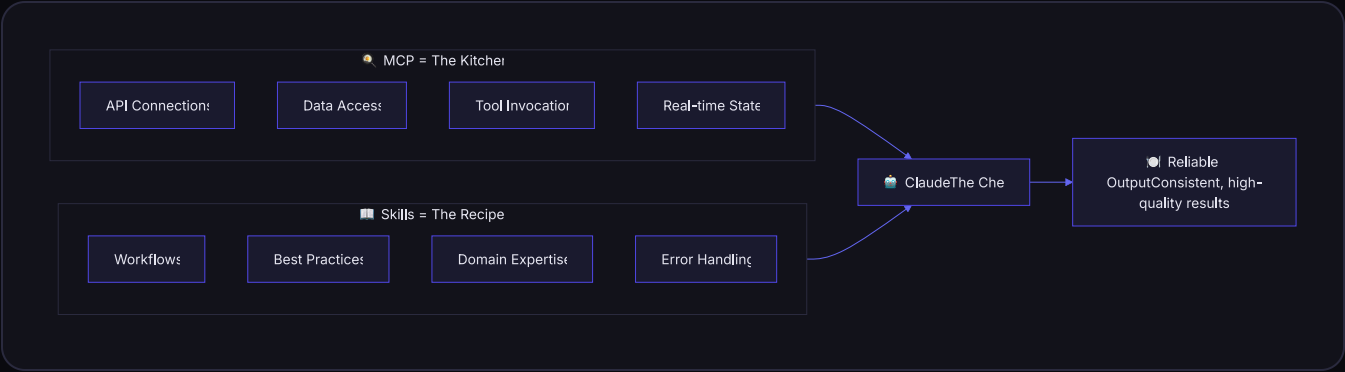
Three-level loading: YAML frontmatter → SKILL.md body → Linked resources.
Hundreds of skills, minimal token overhead.



The MCP + Skills Architecture

1 min

MCP provides capabilities (the kitchen). Skills provide workflows (the recipes). Together: reliable, scalable AI development.

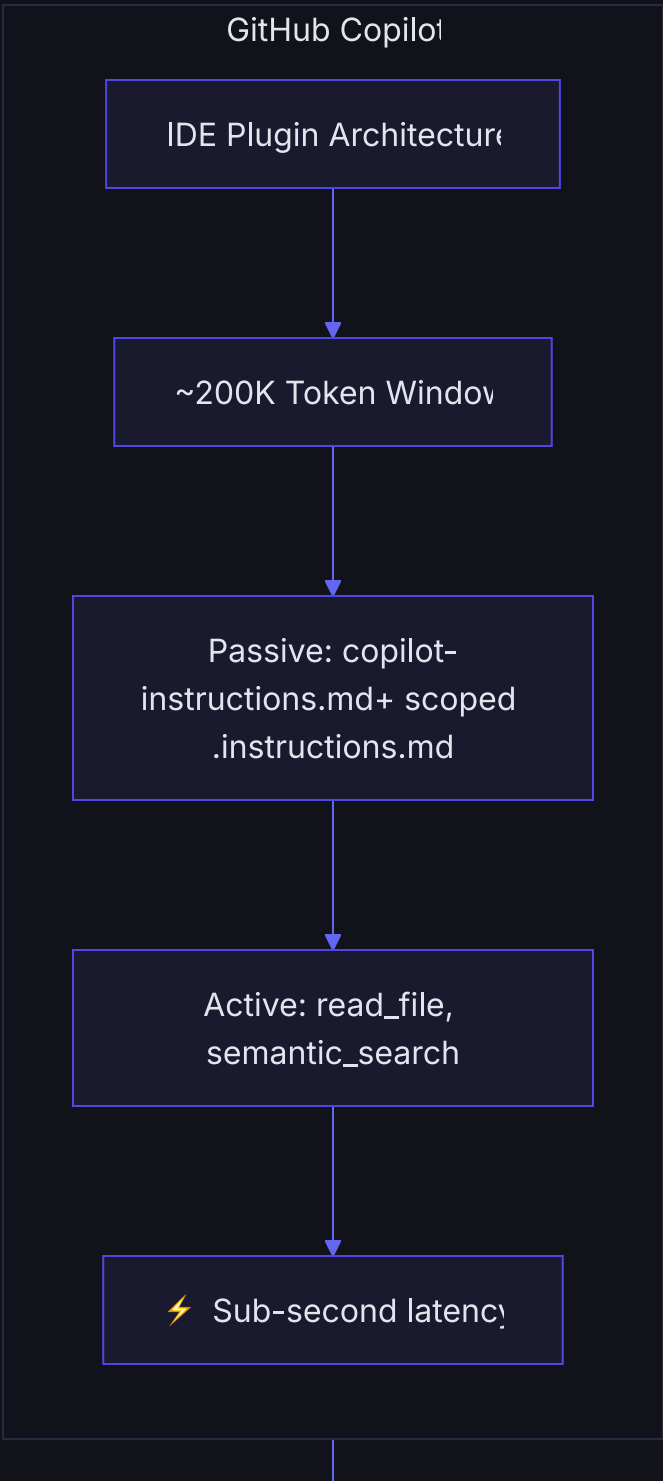


THE PATTERN

Without skills, users connect an MCP and ask *"now what?"* With skills, workflows activate automatically.

Two Architectures 2 min

IDE plugin (sub-second, surgical) vs. autonomous cloud agent (full VM, large-scale). Complementary, not competing.



Dimension	Copilot	Devin
Autonomy	40	95
Setup Speed	95	60
Context Precision	90	85
Large Refactors	50	95
Latency	95	40

The Academic Reality Check

2 min

Vercel shows dramatic gains. ETH Zürich finds marginal effects. The reconciliation reveals when context files actually help.



Key ETH Zürich Finding

✔

Instructions ARE followed
used 1.6x/instance when mentioned

✔

More testing & exploration
Agents search more broadly

⚠

+22% reasoning tokens
Following instructions is harder

✖

LLM-generated files are redundant with existing docs

✔

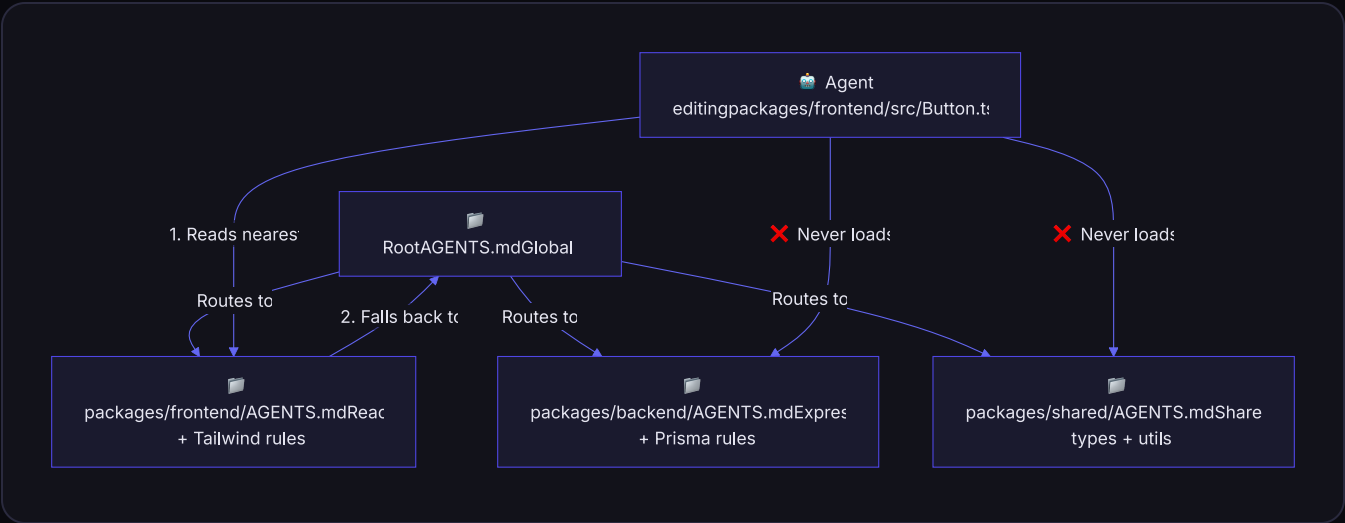
Human-written files provide marginal positive gain

PRACTICAL TAKEAWAY

Don't auto-generate AGENTS.md with an LLM — those are redundant. **Write it yourself.** Encode your *tribal knowledge*.

Nearest-File Precedence1 min

The root AGENTS.md routes. Each package overrides. Zero context leakage between domains.



DESIGN PRINCIPLE

When editing `packages/frontend/src/Button.tsx`, the agent reads **frontend/AGENTS.md** first, then falls back to root. It **never** loads backend or shared rules.

The Hybrid Strategy Stack

1 min

Five layers from foundation (passive, always-present) to large-scale active navigation. Build from the bottom up.

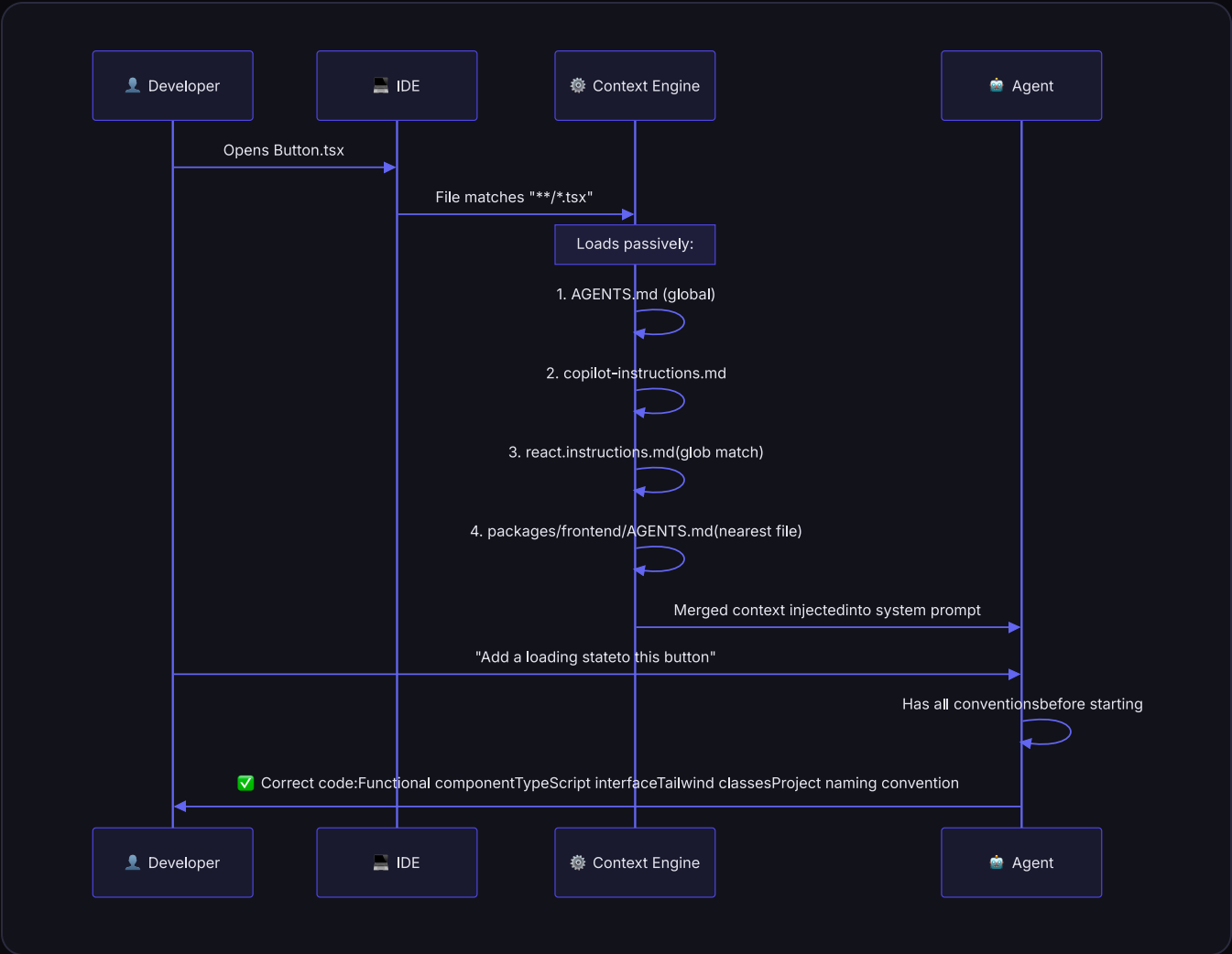


ROI CURVE

Most teams only need the first two layers — **Foundation + Scoped Rules** — to see massive improvements.

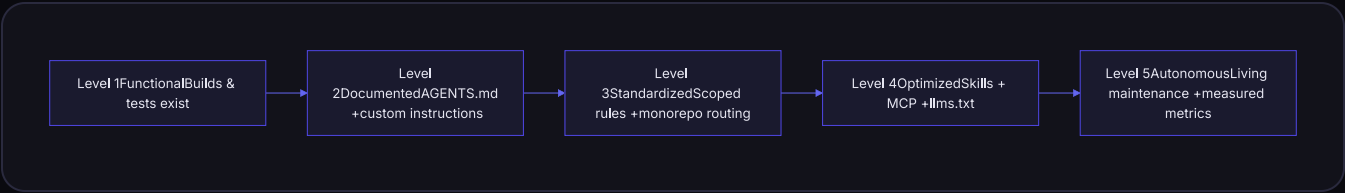
The Passive Steering Workflow 1 min

From file open → glob match → context merge → deterministic output. No searching. No guessing.



Repository Readiness Maturity Model 1 min

Five levels from "it builds" to "fully autonomous." Most teams are at Level 1. Getting to Level 3 takes an afternoon.



L1 → L2

15 minutes

L2 → L3

One afternoon

L3+

Biggest ROI zone

The 15-Minute Audit 2 min

Four phases. Foundation → Scoped Rules → Documentation → Verify. Commit everything.



What Goes in AGENTS.md 1 min

Five sections. Keep under 2 pages. Concise > Comprehensive. 8KB beat 40KB in Vercel's eval.

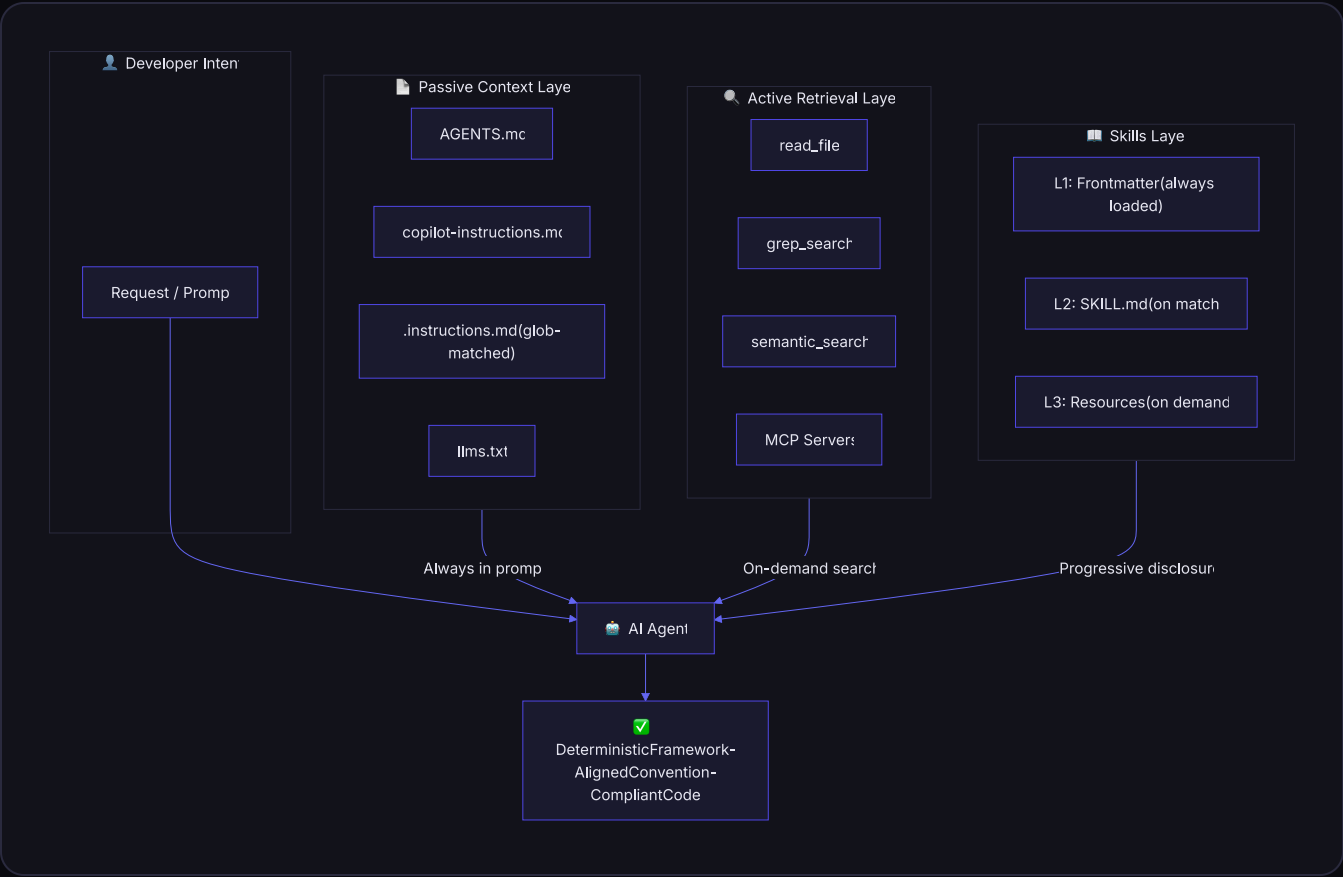


CRITICAL RULE

An 8KB compressed index beat 40KB of full docs. Encode your **tribal knowledge** — the non-obvious conventions, the things that burn people.

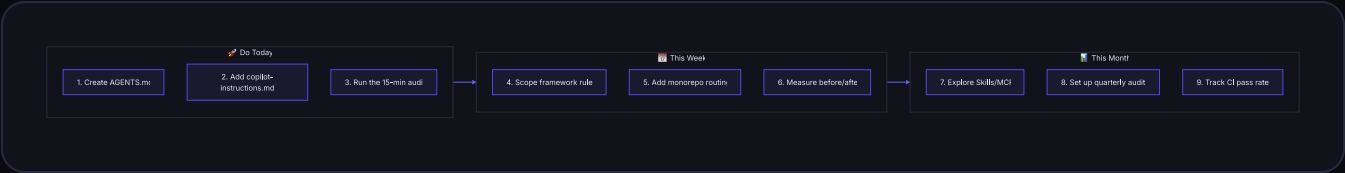
The Complete Context Architecture 1 min

Three layers → one agent → deterministic, framework-aligned, convention-compliant code.



Key Takeaways & Call to Action 2 min

Three time horizons. Today → This Week → This Month.



TODAY — 15 MIN

Create AGENTS.md,
copilot-instructions.md,
run the audit

THIS WEEK

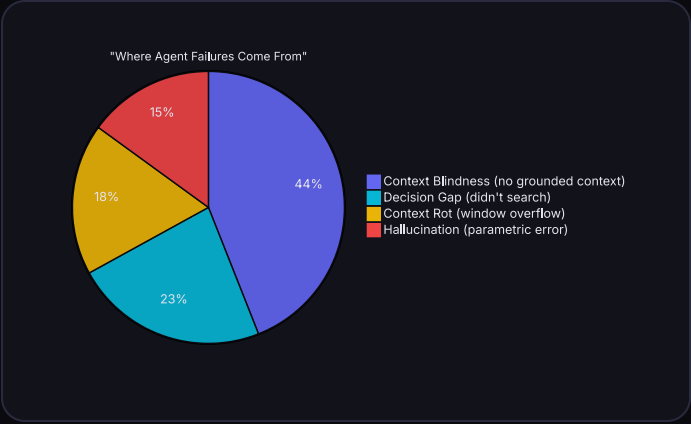
Scope framework rules,
add monorepo routing,
measure before/after

THIS MONTH

Explore Skills/MCP,
quarterly audits, track CI
pass rates

Appendix A: Key Metrics Summary

Where agent failures originate — context blindness dominates at 44%.



44%

Context Blindness — no grounded context provided

23%

Decision Gap — agent didn't search for docs

18%

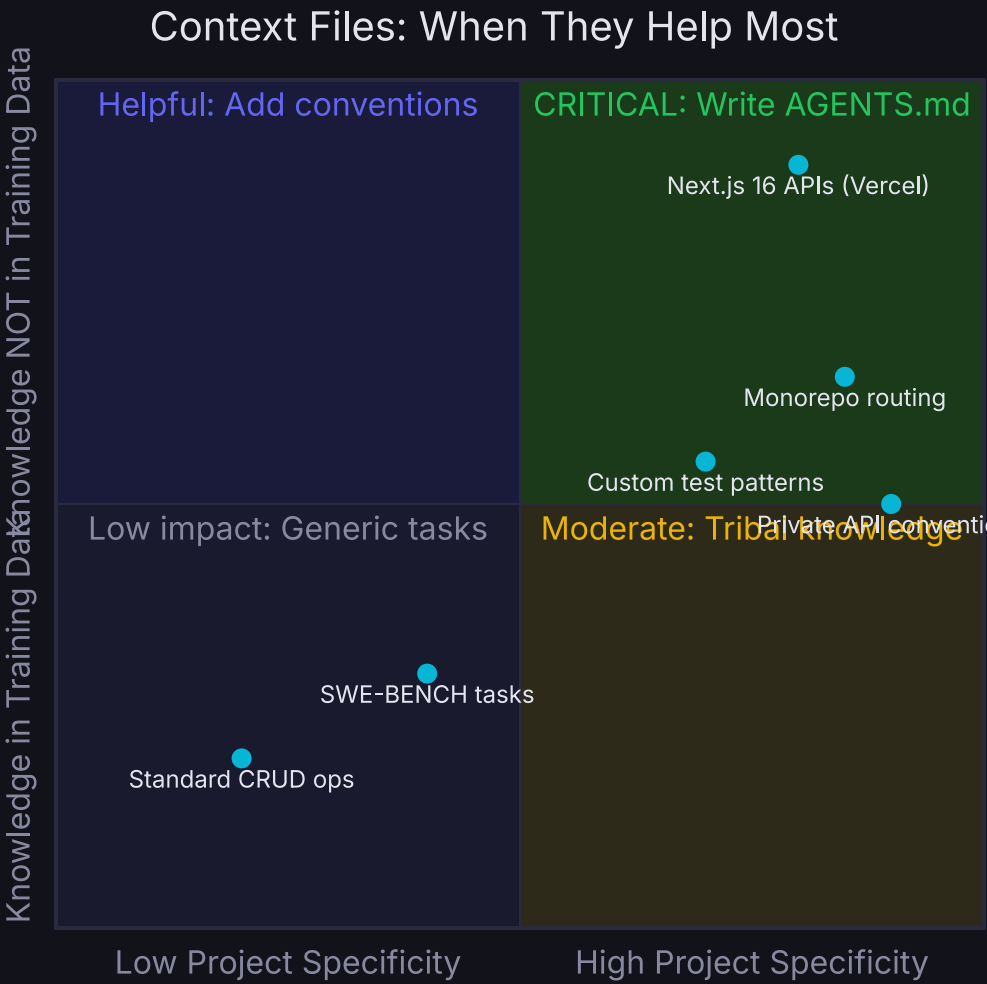
Context Rot — window overflow discards rules

15%

Hallucination — parametric knowledge error

Appendix B: Context File Impact by Scenario

When context files help most: high project specificity + knowledge absent from training data.

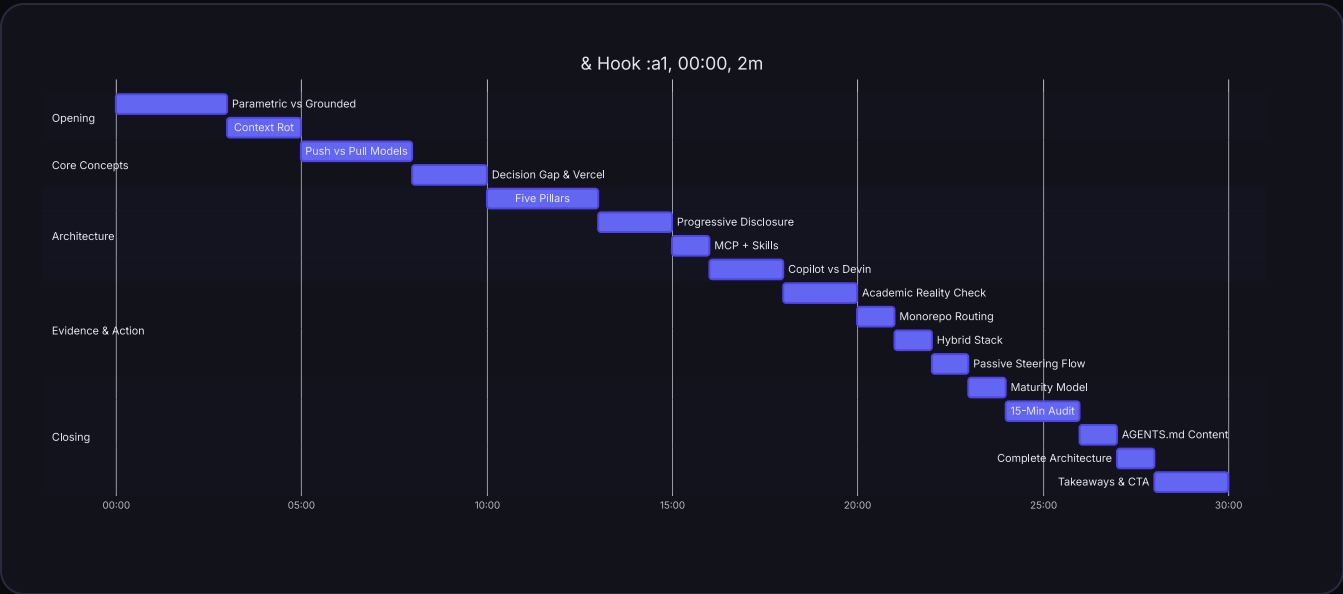


INTERPRETATION

Top-right quadrant = **maximum AGENTS.md impact**. Bottom-left = minimal value. This explains the Vercel vs ETH Zürich discrepancy.

Appendix C: Full Presentation Timeline

20 minutes across 5 sections: Opening, Core Concepts, Architecture, Evidence & Action, Closing.



REFERENCES

Key Citations & Sources

Primary research, specifications, and documentation referenced throughout this talk.

[1] **Vercel Blog** — *AGENTS.md outperforms skills in our agent evals* (Jan 27, 2026)

vercel.com/blog/agents-md-outperforms-skills-in-our-agent-evals

[2] **arXiv 2602.11988** — *Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents?* (ETH Zürich)

arxiv.org/abs/2602.11988

[3] **AGENTS.md Specification** — Official specification for repository-level agent instruction files

agents.md

[4] **GitHub Docs** — *Adding custom instructions for GitHub Copilot*

docs.github.com/.../adding-custom-instructions-for-github-copilot

[5] **Anthropic** — *The Complete Guide to Building Skills for Claude*

platform.claude.com/.../agent-skills/best-practices

[6] **GitHub Blog** — *How to write a great agents.md — Lessons from over 2,500 repositories*

github.blog/.../how-to-write-a-great-agents-md...

[7] **llms.txt Proposal** — Machine-readable documentation standard for LLM consumption

llmstxt.org